

Executive Summary & Recommendation

The **Telco Customer Churn Analysis** project was conducted to identify the key factors contributing to customer churn and provide data-driven recommendations to improve customer retention. The analysis was performed on a dataset containing **7,043 customer records** and **21 business attributes**, including customer demographics, subscription details, internet services, billing information, payment methods, contract types, and customer tenure. Before beginning the analysis, the dataset was cleaned by converting the **TotalCharges** column from string to numeric format, replacing blank values with **0** for customers having zero tenure, and verifying that the dataset contained **0 missing values (0%)** and **0 duplicate records (0%)**, ensuring the reliability and quality of the analysis.

To explore customer behavior and identify churn patterns, multiple **Exploratory Data Analysis (EDA)** techniques were performed using **Python, Pandas, Matplotlib, and Seaborn**. Various visualizations such as **count plots, pie charts, stacked bar charts, histograms, and multiple subplot comparisons** were created to analyze customer demographics, contract types, service subscriptions, payment methods, and tenure. These visualizations helped uncover meaningful business insights and simplified the interpretation of complex customer behavior.

The overall churn analysis reveals that **26.54% (1,869 customers)** have discontinued the company's services, while **73.46% (5,174 customers)** remain active customers. Although the company retains the majority of its customers, losing more than **one out of every four customers** represents a significant business challenge that affects recurring revenue, customer lifetime value, and customer acquisition costs. This indicates that customer retention should be a key strategic focus for the business.

The project utilized a **Count Plot** and **Pie Chart** to analyze the overall churn distribution. The count plot clearly illustrated the number of customers who churned versus those who remained, while the pie chart provided a percentage-wise breakdown of customer retention and attrition. Together, these visualizations offered an immediate understanding of the company's retention performance and highlighted the scale of the churn problem.

Customer demographic analysis was performed using **Count Plots** and **Stacked Percentage Bar Charts**. The gender analysis showed that male and female customers exhibit nearly identical churn behavior, indicating that **gender is not a significant predictor of churn**. However, the stacked percentage chart for **Senior Citizens** revealed a noticeably higher churn percentage compared to non-senior customers. Although Senior Citizens account for only **16.2%** of the customer base, they are more likely to discontinue the service, suggesting the need for personalized customer engagement and support strategies.

A **Histogram** was used to analyze customer tenure. The visualization demonstrated that customers with a tenure of only **1–2 months** are considerably more likely to churn, whereas customers who have remained with the company for a longer period show significantly higher retention. The dataset reports an **average customer tenure of approximately 32.37 months**, indicating that customer loyalty increases over time. This insight emphasizes the

importance of improving customer onboarding and enhancing customer experience during the first few months of service.

Contract type analysis was conducted using a **Count Plot with Churn as the Hue**, which clearly showed that customers on **Month-to-Month contracts** have the highest churn levels compared with customers on **One-Year** and **Two-Year contracts**. Long-term contracts demonstrate significantly stronger customer loyalty, suggesting that encouraging customers to migrate to annual subscription plans through discounts or loyalty programs could substantially reduce churn.

A **3×3 subplot of Count Plots** was developed to compare churn across nine telecom services, including **Phone Service, Multiple Lines, Internet Service, Online Security, Online Backup, Device Protection, Tech Support, Streaming TV, and Streaming Movies**. This consolidated visualization enabled direct comparison across multiple service categories and revealed that customers **without Online Security, Online Backup, Device Protection, and Tech Support** experience significantly higher churn. It also showed that **Fiber Optic Internet** customers have the highest churn among internet service types, while **Phone Service, Streaming TV, and Streaming Movies** have comparatively less influence on customer retention.

Finally, a **Count Plot** was used to evaluate customer payment behavior. The visualization highlighted that customers using **Electronic Check** as their payment method are considerably more likely to churn than customers using **Automatic Bank Transfer, Automatic Credit Card, or Mailed Check**. This finding suggests that encouraging customers to adopt automated payment methods could improve customer convenience and reduce churn.

Key Business Insights

- **Dataset Size:** 7,043 customers with 21 business attributes.
- **Missing Values:** 0 (0%)
- **Duplicate Records:** 0 (0%)
- **Customers Retained:** 5,174 (73.46%)
- **Customers Churned:** 1,869 (26.54%)
- **Senior Citizens:** Approximately 16.2% of the customer base and exhibit a higher churn tendency.
- **Average Customer Tenure:** 32.37 months
- **Highest Churn Risk:** Customers with 1–2 months of tenure.
- **Highest Churn Contract:** Month-to-Month contracts.
- **Highest Churn Internet Service:** Fiber Optic.
- **Highest Churn Payment Method:** Electronic Check.
- Customers without **Online Security, Online Backup, Device Protection, and Tech Support** are significantly more likely to churn.
- **Gender** has minimal influence on customer churn.

Business Recommendations

Based on the analysis, the telecom company should focus on the following strategies to reduce customer churn:

- Encourage customers to upgrade from **Month-to-Month** to **One-Year** or **Two-Year** contracts through discounts and loyalty rewards.
- Strengthen customer engagement during the **first 90 days**, when churn risk is highest.
- Bundle **Online Security**, **Tech Support**, **Online Backup**, and **Device Protection** with internet plans to improve customer retention.
- Investigate the underlying causes of high churn among **Fiber Optic Internet** customers by reviewing pricing, network performance, and customer feedback.
- Promote **Automatic Bank Transfer** and **Credit Card payments** instead of **Electronic Check** by offering incentives for automatic payment enrollment.
- Develop targeted retention campaigns for **Senior Citizens** and newly acquired customers using personalized offers and proactive customer support.
- Implement a predictive churn monitoring system to identify high-risk customers and take preventive action before they cancel their subscriptions.

Conclusion

This project demonstrates how **Exploratory Data Analysis (EDA)** can transform raw customer data into actionable business insights. By leveraging **count plots, pie charts, stacked percentage bar charts, histograms, and multi-panel subplot visualizations**, the analysis successfully identified the major factors driving customer churn. The findings show that **contract type, tenure, internet service, payment method, and value-added support services** are the strongest predictors of churn. Implementing the recommended strategies can help the telecom company reduce churn, improve customer satisfaction, increase customer lifetime value, and drive sustainable business growth. This project also showcases practical data analysis skills in **Python, Pandas, Matplotlib, and Seaborn**, demonstrating the ability to convert business data into meaningful, data-driven decisions.